

## **Overview**

Throughout our experimentation, different models exhibited superior performance depending on the specific classification targets or generalization conditions. However, the model that demonstrated the most consistent balance between classification accuracy and generalizability was a fine-tuned version of the Whisper Large encoder, which we used as our final model. Whisper Large (WhL) is a multilingual transformer-based encoder-decoder architecture originally developed for automatic speech recognition. In our cognitive state classification pipeline, we used the encoder component of WhL followed by a classification head comprised of a multi-layer perceptron. The model processes raw audio by converting it into log-Mel spectrograms, which are then passed through a stack of transformer encoder blocks. To accommodate the limited size of our training dataset, we applied a selective layer freezing strategy. Specifically, we only allowed the first two and the final encoder layers to remain trainable, preserving much of the pretrained knowledge while enabling task-specific learning.

Model training was conducted using data from 12 studies within the DementiaBank repository. These included Baycrest, Delaware, Holland, Kempler, Lanzi, Lu (English), Pitt, WLS, DePaul, Hopkins, Jalvingh, and Ivanova. For recordings with multiple speakers, we utilized time-aligned transcripts to isolate participant speech, excluding segments of other individuals. This preprocessing ensured that only participant speech was retained and subsequently trimmed for consistency. All recordings underwent a voice enhancement pipeline designed to improve speech clarity and reduce variability in acoustic quality. This pipeline included background noise reduction, dynamic range compression, frequency equalization, and loudness normalization. Once processed, the recordings were segmented into 30-second chunks to match the maximum input size of the Whisper’s encoder. The final prediction for each participant was computed as an average of the class probabilities generated from their individual audio segments.

The classification task was structured around four diagnostic categories: (i) healthy controls (HC; 2044 samples), (ii) mild cognitive impairment (MCI; 559 samples), (iii) Alzheimer’s disease (AD; 946 samples), and (iv) other dementias (OD; 171 samples). The last class comprises participants with the following diagnoses across all the studies: dementia, vascular, PPA-nos, LVPPA, SVPPA, NFAPPA, FTD, other, semantic PPA, Pick’s, aphasia, SD, LPA, and PNFA. The number of samples for each class shows a high imbalance in the data which imposes challenges for AI development. The initial model evaluation was conducted using four distinct test sets: (i) the Phase 2 Eureka challenge test set, (ii) a study with an unseen task (VAS) to assess out-of-distribution task generalization, (iii) a set containing Mandarin recordings (Chou, Lu, Ye), representing an unseen language, and (iv) a composite set designed to preserve an approximate 80/20 train-test split, including samples from (i) to (iii) plus additional held-out recordings.

To further investigate the generalizability of the best performing model, we developed a more comprehensive evaluation framework based on two complementary strategies. The first strategy involved the design of a generalization benchmark, comprised of two leave-one-out protocols. One protocol focused on language robustness by training three separate models, each excluding one of the languages present in the dataset (English, Spanish, or Mandarin). The performance of each model was then evaluated on the held-out language to assess the model’s ability to generalize to entirely unseen language during fine-tuning. The second protocol was designed to evaluate generalization across cognitive tasks. Since recordings may contain overlapping task elements and task-specific segmentation is not automated, we manually curated a subset of recordings where a single task was predominantly featured. We focused on three representative tasks: Image Description, Word Fluency, and Discourse. In some cases, the discourse task overlaps with the two other tasks, however, most of the content related to the held-out task was removed during training. Separate models were then trained using a leave-one-task-out approach, where all recordings primarily related to a selected task were excluded during training. This design enables us to assess the contribution of each task to model performance and identify which tasks are most sensitive to omission during training.

The second strategy specifically addresses the language imbalance present in existing datasets, which are predominantly composed of English-language recordings. To mitigate this bias while taking advantage of the strong performance of models trained on English data, we developed a speech translation pipeline. This pipeline is designed to convert non-English recordings into English while preserving key acoustic characteristics of the original speakers. This enables the prediction of cognitive states for non-

English-speaking individuals using models trained exclusively or predominantly on English data, which remains the most abundant and accessible linguistic resource in this domain.

### **Methodology and Performance**

We propose a set of evaluation metrics to perform a more rigorous and comprehensive assessment of the model. To move beyond the limitations of log loss alone, we incorporate accuracy, precision-recall (PR) curves, and area under the PR curves (AUPRC). PR curves offer a more informative view in class-imbalanced scenarios, allowing us to evaluate the trade-off between sensitivity (recall) and the ability to avoid false positives (precision). This information is critical in AD/ADRD screening, where the cost of misclassification varies by clinical context and diagnosis stage. Our Phase 3 experimentation focused on understanding how model architecture, input length, and auxiliary information affect prediction.

One key finding concerns the amount of audio needed for reliable classification. When using the distilled Whisper model (a smaller version optimized for efficiency) trained only on the first 30 seconds of each recording, it outperformed training on the full recordings, even when using segment-level ensemble. In contrast, the larger WhL encoder significantly benefited from full-audio training. The AUPRC improved from 0.541 to 0.611, and specifically for heterogeneous OD group it increased approximately two-fold.. These results suggest that while smaller models may overfit when exposed to more data, larger models are better equipped to extract meaningful information from longer recordings. This underscores the importance of correct model selection based on data availability. Additionally, when training data are limited, careful optimization remains essential. We found that freezing all but the first two and the final encoder layers in WhL helped retain the benefits of pretraining knowledge while allowing for task-specific learning. This strategy led to the best overall performance across diagnostic groups, without the computational burden of retraining the entire model.

Given the particular characteristics of speech in individuals with dementia, we also evaluated CrisperWhisper, a variant of the Whisper model that incorporates special markers for repetitions, partial words, and filler expressions; features commonly associated with neurodegenerative speech patterns. Despite its theoretical relevance, it yielded only marginal improvements for AD and led to decreased performance for HC, MCI, and OD. As discussed on the contributing factors section, our results suggest that WhL, when fine-tuned on this specific dataset, is already learning to capture the dynamics of neurodegeneration-related speech. This may explain why the additional cues introduced by CrisperWhisper did not translate into the substantial performance gains that might otherwise be expected.

Furthermore, given the relevance of known risk factors such as sex, age, and level of education in dementia, we explored how incorporating this demographic information affected model performance. Adding age alone notably increased the AUPRC for OD, from 0.399 to 0.735, indicating potential utility for certain clinical subtypes. However, for other diagnostic categories, the integration of demographic metadata had minimal impact. When age was included with a learnable scaling factor, the model quickly learned to ignore it, suggesting the information was redundant with what the model had already inferred from the audio. These findings suggest that demographic features can be valuable when they provide clinically relevant variation not already captured by acoustic patterns but may be unnecessary in cases where the speech signal contains sufficient discriminatory information.

To better capture the inherent sequential nature of speech and leverage the longer recordings available in some cases, we implemented a two-stage modeling framework. In this architecture, Whisper embeddings from successive segments were passed to a lightweight transformer that modeled relationships across time. This approach, on 30-second segments, achieved the highest performance for HC and, when using shorter 10-second segments to obtain more granularity and context within the sequence, also produced the best performance for OD. Notably, this group had a greater proportion of longer recordings compared to other diagnostic categories, which typically included only one to three 30-second segments per participant. This likely contributed to the transformer's effectiveness in capturing longer-range speech dependencies for OD, where more temporal context was available. These results highlight the value of modeling speech patterns over time, particularly for conditions where longer and more complex speech samples are present.

We also evaluated the inclusion of SMILE acoustic features, but this consistently reduced performance. This decline is likely related to the fact Whisper’s embeddings already capture rich spectral and prosodic information, making the fusion of hand-crafted features redundant or even disruptive. Similarly, we applied class-weighting to mitigate label imbalance by assigning greater penalty to misclassifications in underrepresented classes. This strategy did not yield meaningful improvements. These results highlight the limitations of post hoc algorithmic corrections and instead underscore the importance of collecting larger, more balanced datasets to improve model robustness across diagnostic groups.

Following all experimentation, we observed that the best-performing model varied by diagnostic category. For instance, the highest performance for HC was achieved with the two-stage transformer on 30-second segments. For MCI and AD cases, Distilled Whisper outperformed other models. For other types of dementia, the two-stage transformer on 10-second segments was most effective. This suggests that future clinical tools may benefit from diagnosis-specific models or ensembles, rather than relying on a single, universal classifier. Nevertheless, the model with the strongest overall performance across all classes and better generalization to translated audios, as shown below, was WhL. The class-specific performance of the best model, in comparison with other top-performing models, is presented in Figure 1.

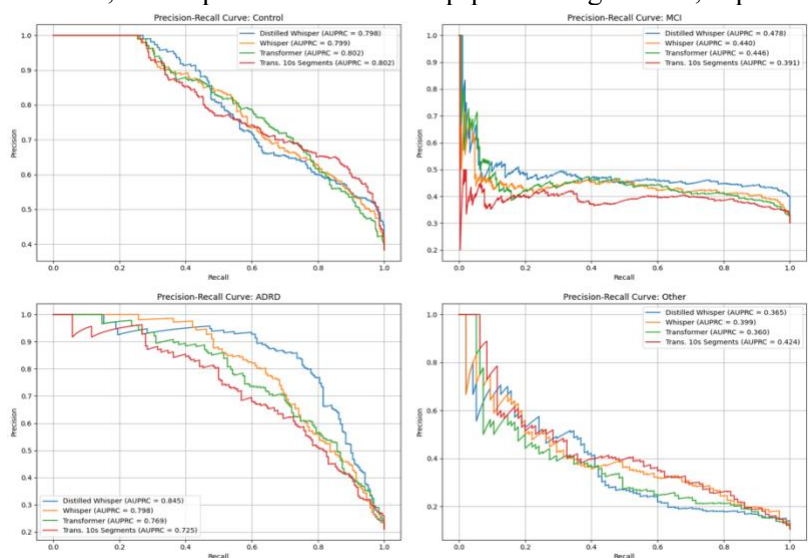


Figure 1. Precision-Recall curves per class in the top performing models.

multi-task learning approaches could enable the model to more effectively distinguish between clinically similar but diagnostically distinct groups. Our results also suggest that temporal modeling contributes significantly to improved classification of the "OD" group, likely due to the availability of longer recordings for these participants. This highlights the benefit of modeling speech over time and suggests that incorporating more granular, segment-level attention mechanisms may help further. Data augmentation techniques or contrastive learning strategies could also improve representation learning for minority classes. Additionally, identifying which cognitive tasks yield the most informative speech samples for each condition, and incorporating task type as a modeling input, could allow for more tailored predictions. Finally, as shown below, audios translation from less-represented languages to English also improve predictive performance. In summary, our results show that larger pretrained models, careful adaptation via partial fine-tuning, and modeling speech patterns over time are key to improving the accuracy and reliability of speech-based AD/ADRD prediction. Demographic information may offer selective value, but additional hand-crafted features and algorithmic balancing techniques provided limited benefit. Depending on the specific goals of individual researchers, our findings suggest multiple viable directions for future development. These potential next steps are further outlined in the Data section.

In our evaluation of model generalizability through speech-to-speech (STS) translation, diagnostic prediction improved across all non-HC classes within the Mandarin-language test subset. This outcome was anticipated, as Mandarin was not present in the training corpus, and translation to English allowed the model

The worst performing classes were MCI and OD, which is expected given that both are the least represented in the dataset. Improving performance for these groups will likely require a combination of more targeted data collection and refined modeling strategies. Expanding the dataset with additional recordings from individuals with MCI and less common dementia subtypes would help the model better learn the diverse speech patterns associated with these conditions. At the same time, diagnosis-specific models or

to leverage learned representations from the dominant training language. Specifically, diagnostic accuracy for HC reduced 10% (from 0.53 to 0.48), while it increased 82% in AD (from 0.50 to 0.92), and slightly increased for MCI (from 0.50 to 0.51) and OD (from 0.83 to 0.86). These improvements suggest that converting non-English speech to English, while preserving speaker characteristics, can effectively reduce the language domain gap and improve model performance. Interestingly, translation also led to a substantial increase in accuracy for Spanish-speaking participants with HC and AD rising from 0.40 to 0.56 ( $\approx 41\%$  improvement), and from 0.68 to 0.88 ( $\approx 30\%$  improvement) respectively, despite Spanish being present in the training data. Additionally, for the other diagnostic groups in the Spanish subset, the performance was kept almost the same. However, precision for the AD cases significantly declined (from 0.4545 to 0.1818). This decline may be attributed to translation artifacts and the subtle speech markers typical of AD, such as pauses, hesitations, and disfluencies. These features are susceptible to being smoothed over or distorted during automatic translation, particularly by open-source models not optimized for clinical usage. These findings underscore both the potential and limitations of STS translation as a strategy for improving generalization across languages. While state-of-the-art, closed-source models have made significant advances in high-fidelity translation, current open-source tools still fall short in preserving the fine-grained acoustic and linguistic cues essential for detecting neurodegenerative speech patterns. In contexts where training data for a target language is limited, translation may nonetheless serve as a valuable interim approach, especially as translation quality continues to improve, by enabling the use of well-trained English models in broader multilingual settings.

### **Data**

Out of distribution experiments provided insights into which tasks and languages aid the model to generalize better. By removing English data, the model showed the worst performance overall, which is expected because it's the biggest corpus and consequently the biggest sample of different speech changes between diagnosis classes. After English, the next task that affected the performance the most is image description. This task mostly comprises the cookie theft picture task, all in English, and some mandarin samples. On the other hand, removing discourse and word fluency tasks didn't affect the classification as much as in the other experiments. Efforts should be targeted to develop a more robust analysis to identify granular tasks that improve and decrease generalizability.

Our findings emphasize that data-related decisions, both in terms of recording design and linguistic diversity, play a critical role in model performance and generalizability. Notably, training smaller models on just the first 30 seconds of each audio consistently outperformed using full recordings. This suggests that collecting a larger number of short, diverse recordings may be more beneficial than acquiring fewer, longer sessions, particularly for lightweight models and large-scale screening applications. However, higher capacity models such as WhL and the two-stage transformer showed clear gains from full-length audio, particularly in capturing sequential features relevant for complex cases like OD. This points to two viable strategies for future data collection: maximizing variability through short recordings when using small models or leveraging longer recordings to enrich sequential speech modeling in high-resource settings. In parallel, our evaluation of STS translation demonstrated promising gains for underrepresented languages. Translating Mandarin audio into English significantly improved classification accuracy across all non-HC categories, and translation also improved detection of HC, MCI, and AD in Spanish indicating potential benefits for underrepresented languages. These results highlight the dual importance of both increasing language coverage in training data and improving translation fidelity, particularly for clinical features that are acoustically nuanced. These insights suggest that future data efforts should be tailored to the research context, whether optimizing for coverage, linguistic diversity, or model architecture, and that speech translation may serve as a valuable complementary strategy when native-language training data are limited.

### **Contributing factors**

To understand which acoustic features the model uses for cognitive states classification, we conducted neural probing analysis across the entire dataset. Neural probing trains simple linear classifiers to predict specific features from a model's internal representations, according to Bengio et al. if a probe can accurately predict an acoustic feature (or any other objective such as the class) from neural activations, this indicates the model has preserved that information during processing<sup>1</sup>. We extracted representations from 9

transformer layers and trained logistic regression probes to predict 57 acoustic features organized into 9 categories. Each feature was converted to binary classification problem (above/below median) and evaluated using 5-fold cross-validation. Performance above 50% (chance level) indicates the model encodes that acoustic information.

Energy & Amplitude relies on Root Mean Square (RMS) energy analysis and represents vocal intensity and temporal energy distribution. Spectral Shape features describe the frequency distribution of speech energy. The centroid indicates the perceived brightness, bandwidth measures spectral spread, rolloff shows high-frequency energy distribution, contrast captures the harmonic structure, and flatness measures tonality versus noise. Spectral Dynamics quantify how quickly spectral characteristics change over time. Pitch & F0 features capture the fundamental frequency patterns including central tendency, variability, voicing patterns, and local perturbations (jitter). Voicing & ZCR analysis distinguishes voiced from unvoiced speech segments through zero-crossing rate patterns. MFCC features (Mel Frequency Cepstral Coefficients) provide spectral representations. Temporal & Rhythm features measure speech timing through onset detection and rhythm consistency. Pauses & Silence patterns indicate fluency and language processing difficulties through pause frequency and silence proportion analysis.

| Feature Types      | Features.                                                                                                                                              |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Energy & Amplitude | RMS: mean, standard deviation, range, skewness, kurtosis. Peak-to-RMS ratio. Segment analysis: 3 temporal segments. Energy variability across segments |
| Spectral Shape     | Centroid, bandwidth, rolloff, contrast, flatness                                                                                                       |
| Spectral Dynamics  | Flux and stability(frame-to-frame correlation)                                                                                                         |
| Pitch & F0         | F0, F0 contour, F0 jitter                                                                                                                              |
| Voice Quality      | Harmonic-to-noise ratio: approximation. Shimmer: approximation                                                                                         |
| Voicing & ZCR      | Zero-crossing rate, ZCR segments, and ZCR variability across segments                                                                                  |
| MFCC               | MFCC coefficients 1-6: mean, standard deviation. MFCC deltas 1-3: mean temporal derivatives                                                            |
| Temporal & Rhythm  | Speech rate: onsets per second. Rhythm regularity: inter-onset interval stability                                                                      |
| Pauses & Silence   | Pause rate: approximation. Silence ratio: proportion of silent frames                                                                                  |

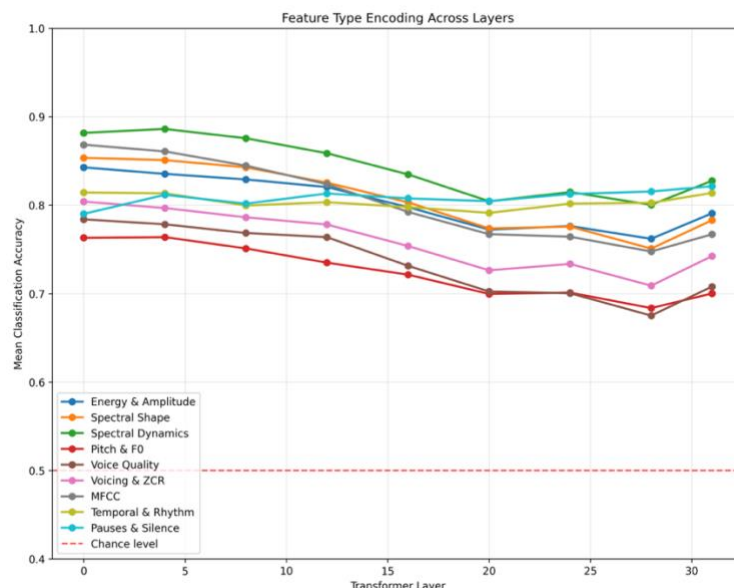


Figure 2. Layer-Wise Feature Type Encoding Accuracy

Spectral Dynamics, Pauses & Silence and Temporal & Rhythm are the top 3 feature types encoded by whisper last layers, indicating that these types of features are the most contributing factors on the model ability to classify cognitive states from speech representation. The prominence of Pauses & Silence type of features is particularly significant as previous research has shown pause patterns in speech are useful

Figure 1 summarizes the layer-wise aggregated by feature type accuracy of the acoustic probing. In general, these results illustrate how the whisper model learns the underlying speech structures without explicitly being trained on them. Considering that this technique was applied to the WhL fine-tuned model we focus our analysis on the first and last 2 layers because the other intermediate layers were frozen. The results obtained in the frozen layers demonstrate Whisper's inherent capacity to preserve relevant acoustic information, maintaining 70-85% accuracy. This indicates that general speech recognition models capture fundamental acoustic patterns that transfer effectively to this application.

markers to detect dementia<sup>2-4</sup>. Additionally, Temporal & Rhythm features in particular, speech rate and rhythm regularity have been documented to be altered in AD patients in contrast to HC<sup>5</sup>.

## References

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